

Customer Shopping Behaviour Analysis

An end-to-end analysis of 3,900 customer purchases using Python, SQL and Power BI, aimed at understanding who our customers are, what they buy, and how to grow revenue.

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Prepared by Nigar Shahmuradova · Data Analyst · October 2026

Tools: Python (Pandas, Matplotlib) · SQL Server · Power BI

1 Project Overview

A retail company wants to understand how its customers shop: which products sell best, which customer groups spend the most, and how things like discounts, subscriptions, shipping and seasons relate to buying. The goal of this project is to turn raw sales data into clear findings the business can act on.

Business questions

- Who are our customers (age, gender, location) and how much do they spend?
- Which categories and products bring in the most revenue and the best reviews?
- Do discounts, subscriptions and faster shipping make customers spend more?
- How many customers are new, returning or loyal, and how can we keep them?

Workflow

1 · Collect Load the CSV file (3,900 rows) into Python.	2 · Clean Fix missing values, rename columns, add new helper columns.	3 · Explore Summary statistics and charts in Python.	4 · Query Answer 10 business questions with SQL.	5 · Visualise Interactive Power BI dashboard.	6 · Recommend Turn findings into business actions.
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Headline numbers

3,900 Customers / purchases	\$233,081 Total revenue	\$59.76 Average purchase
3.75 / 5 Average review rating	25.35 Average previous purchases	89 days Average time between purchases

Key takeaway in one sentence: revenue is spread evenly across seasons, payment methods and age groups, so the biggest growth levers are **getting women to subscribe** (today 0% do), **pushing Clothing and Accessories** (77% of revenue), and **rewarding the large loyal base** (80% of customers have 10+ previous purchases).

2 Dataset Summary

The dataset has **3,900 rows** (one purchase per customer) and **18 columns**. Only one column had missing values: **Review Rating** (37 empty cells, under 1%). There were no duplicate customers.

Column	Type	What it means / values
Customer ID	Number	Unique ID, 1 – 3,900
Age	Number	18 – 70 years, average 44
Gender	Text	Male 2,652 (68%) · Female 1,248 (32%)
Item Purchased	Text	25 products (Blouse, Jewelry, Pants, Shirt ...)
Category	Text	Clothing 1,737 · Accessories 1,240 · Footwear 599 · Outerwear 324
Purchase Amount (USD)	Number	\$20 – \$100, average \$59.76
Location	Text	50 US states
Size · Color	Text	Sizes S / M / L / XL (M is most common) · 25 colours
Season	Text	Spring 999 · Fall 975 · Winter 971 · Summer 955
Review Rating	Decimal	2.5 – 5.0, average 3.75 · 37 missing
Subscription Status	Yes / No	Yes 1,053 (27%) · No 2,847
Shipping Type	Text	6 options (Free, Standard, Store Pickup, Next Day Air, Express, 2-Day)
Discount Applied	Yes / No	Yes 1,677 (43%)
Promo Code Used	Yes / No	Exactly the same values as Discount Applied → removed
Previous Purchases	Number	1 – 50, average 25.35
Payment Method	Text	6 options, each used by roughly 16% of customers
Frequency of Purchases	Text	Weekly → Annually (7 options)

Columns added during cleaning

New column	How it was made
age_group	Age split into 4 equal-size groups: young-adult, adult, middle_aged, senior
purchase_frequency_days	Frequency turned into days (Weekly = 7, Bi-Weekly/Fortnightly = 14, Monthly = 30, Quarterly/Every 3 Months = 90, Annually = 365)

3 Exploratory Data Analysis (Python)

3.1 Data loading and cleaning

```
# 1. Load and inspect
import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')
df.info(); df.describe(include='all'); df.isnull().sum()    # Review Rating: 37 missing

# 2. Fill missing ratings with the median rating of the same category
df['Review Rating'] = df.groupby('Category')['Review Rating'] \
    .transform(lambda x: x.fillna(x.median()))

# 3. Column names to snake_case
df.columns = df.columns.str.lower().str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

# 4. New column: age groups (4 equal-size groups)
df['age_group'] = pd.qcut(df['age'], q=4,
    labels=['young-adult', 'adult', 'middle_aged', 'senior'])

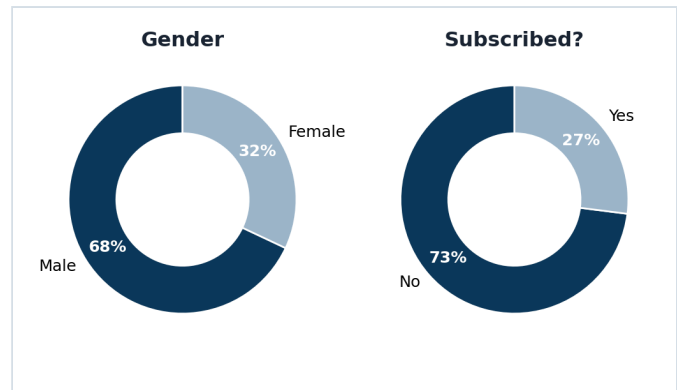
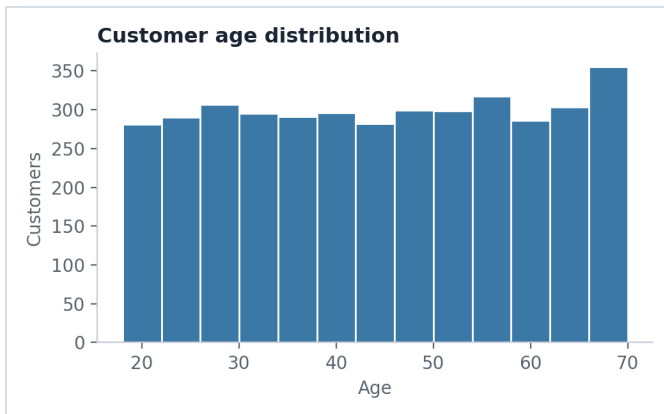
# 5. New column: purchase frequency in days
freq = {'Weekly': 7, 'Bi-Weekly': 14, 'Fortnightly': 14, 'Monthly': 30,
    'Quarterly': 90, 'Every 3 Months': 90, 'Annually': 365}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(freq)

# 6. Remove the duplicate column (promo_code_used = discount_applied in every row)
(df['discount_applied'] == df['promo_code_used']).all()    # True
df = df.drop('promo_code_used', axis=1)

# 7. Load the clean table into SQL Server for analysis
from sqlalchemy import create_engine
engine = create_engine('mssql+pyodbc://.../customer_behavior?driver=ODBC+Driver+17+for+SQL+Server')
df.to_sql('customer', engine, if_exists='replace', index=False)
```

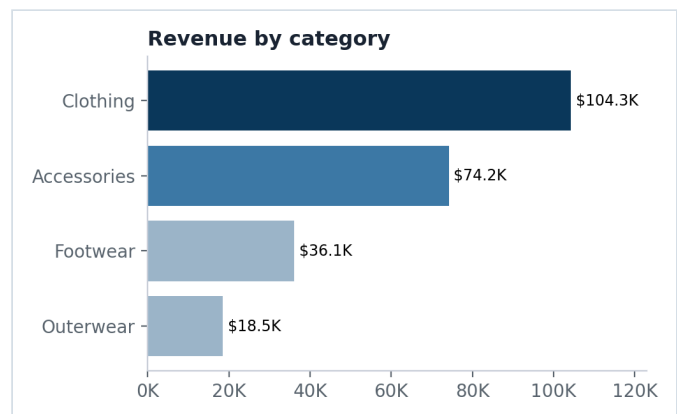
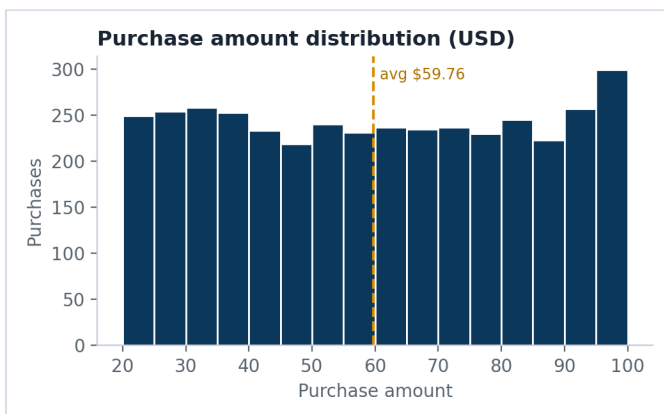
Why the category median? Ratings differ a little by category, so using each category's own middle value is fairer than one overall average, and the median is not pulled by extreme values.

3.2 Who are the customers?



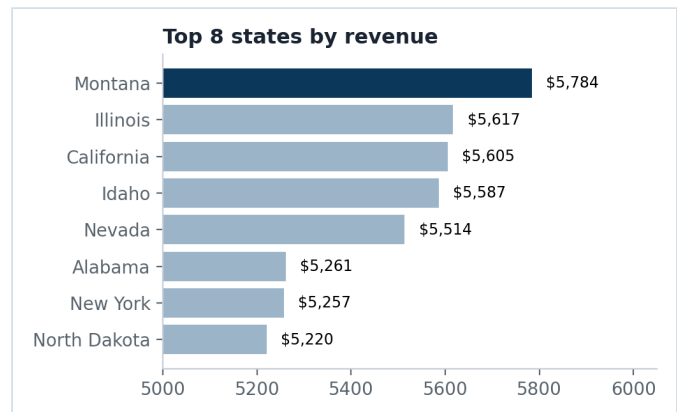
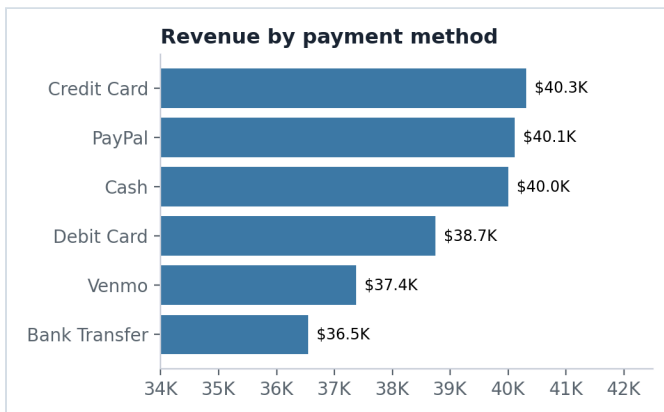
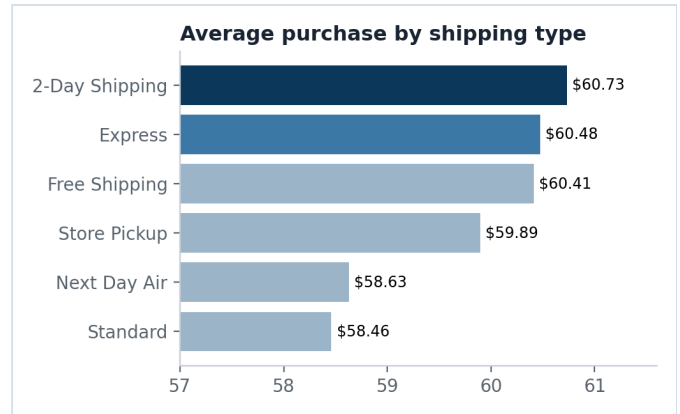
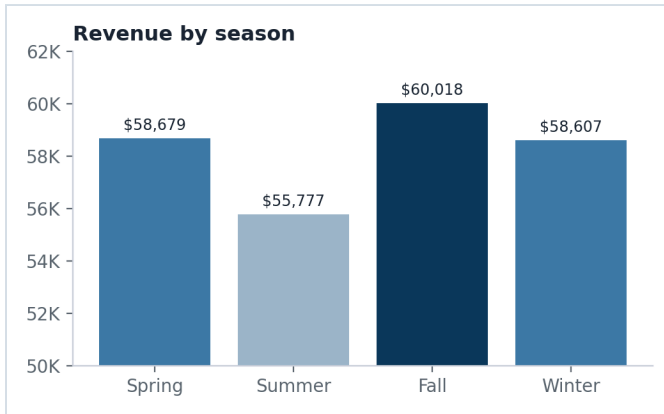
- **Age is spread evenly** from 18 to 70 – there is no single “typical” age; each age group holds about a quarter of customers.
- **68% of customers are men.** They also bring in 68% of revenue (\$157.9K vs \$75.2K).
- **Only 27% are subscribers - and all 1,053 subscribers are men.** Not a single female customer is subscribed. This is the clearest gap in the data.

3.3 What and how much do they buy?



- Purchase amounts are spread evenly between \$20 and \$100 (average **\$59.76**), with no extreme outliers.
- **Clothing (\$104.3K) and Accessories (\$74.2K) make up 77% of revenue.** Outerwear brings in only \$18.5K (8%).

3.4 When, where and how do they buy?



- **Fall is the best season** (\$60.0K) and **Summer the weakest** (\$55.8K) – a gap of about 7%.
- Customers who choose **2-Day, Express or Free shipping** spend about **\$2 more per order** than those who choose Standard or Next Day Air.
- All six payment methods are used almost equally (Credit Card and PayPal slightly ahead).
- **Montana, Illinois and California** are the top states by revenue, but the difference between states is small.
- Customers with a discount spend about the same as those without (\$59.28 vs \$60.13). **Discounts are not increasing order size.**

4 SQL Business Analysis

The clean table customer was loaded into SQL Server and 10 business questions were answered. Each question shows the query, its result and what it means.

Q1 How much revenue do men and women bring in?

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

gender	revenue
Male	157,890
Female	75,191

Men bring in about 2× more revenue than women, mainly because there are 2× more male customers.

Q2 Which customers used a discount but still spent more than average?

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount)
                        FROM customer);
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
... first 5 rows shown	

A large group of discount users still buy above-average orders – these are good targets for “spend more, save more” offers.

Q3 Which 5 products have the highest average review rating?

```
SELECT TOP 5 item_purchased,
ROUND(AVG(CAST(review_rating AS DECIMAL(10,2))),
2)
AS average_rating
FROM customer
GROUP BY item_purchased
ORDER BY average_rating DESC;
```

item_purchased	average_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Best-rated items are mostly footwear and accessories, but even the top rating is below 3.9 – there is room to improve satisfaction overall.

Q4 Do Express customers spend more than Standard customers?

```
SELECT shipping_type,
       AVG(purchase_amount) AS average_purchase
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

shipping_type	average_purchase
Express	60
Standard	58

Express customers spend about \$2 (3.5%) more per order – faster delivery is linked with bigger baskets.

Q5 Do subscribers spend more than non-subscribers?

```
SELECT subscription_status,
       COUNT(customer_id) AS customers,
       AVG(purchase_amount) AS avg_spend,
       SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

status	customers	avg_spend	total_revenue
Yes	1,053	59	62,645
No	2,847	59	170,436

Subscribers do not spend more per order. The subscription programme currently gives no extra value per purchase.

Q6 Which 5 products are most often bought with a discount?

```
SELECT TOP 5 item_purchased,
       SUM(CASE WHEN discount_applied = 'Yes'
                THEN 1 ELSE 0 END) * 100 / COUNT(*)
       AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC;
```

item_purchased	discount_rate %
Hat	50
Coat	49
Sneakers	49
Sweater	48
Pants	47

About half of all Hats, Coats and Sneakers are sold at a discount – these items may depend too much on price cuts.

Q7 How many customers are New, Returning and Loyal?

```
WITH customer_type AS (
  SELECT customer_id,
         CASE WHEN previous_purchases <= 2 THEN 'New'
              WHEN previous_purchases <= 10 THEN 'Returning'
              ELSE 'Loyal' END AS customer_segment
  FROM customer)
SELECT customer_segment,
       COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment;
```

customer_segment	customers
New	155
Returning	629
Loyal	3,116

80% of customers are loyal (more than 10 previous purchases), while only 4% are new. The business keeps customers well but attracts few new ones.

Q8 What are the top 3 products in each category?

```
WITH item_counts AS (  
  SELECT category, item_purchased,  
         COUNT(customer_id) AS total_orders,  
         ROW_NUMBER() OVER (PARTITION BY category  
                             ORDER BY COUNT(customer_id) DESC) AS dr  
  FROM customer  
  GROUP BY category, item_purchased)  
SELECT dr, category, item_purchased, total_orders  
FROM item_counts  
WHERE dr <= 3;
```

dr	category	item	orders
1	Accessories	Jewelry	171
2	Accessories	Belt	161
3	Accessories	Sunglasses	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Jewelry, Blouse, Pants, Sandals and Jacket are the “hero” products of their categories – they should always be in stock and front of the store.

Q9 Are repeat buyers (5+ previous purchases) more likely to subscribe?

```
SELECT subscription_status,  
       COUNT(customer_id) AS customers  
FROM customer  
WHERE previous_purchases >= 5  
GROUP BY subscription_status;
```

subscription_status	customers
Yes	980
No	2,583

Even among repeat buyers only 28% subscribe – almost the same as the overall rate. 2,583 loyal non-subscribers are the easiest people to convert.

Q10 Which age group brings in the most revenue?

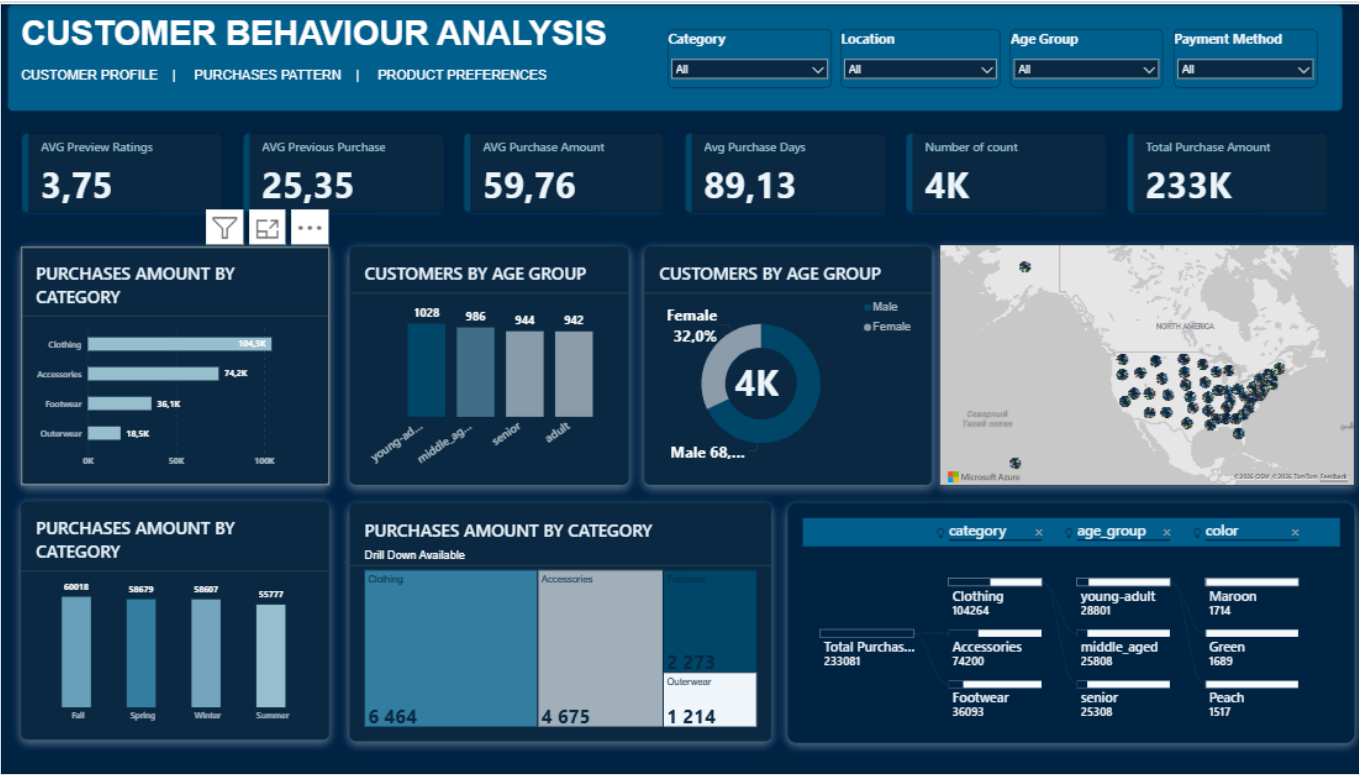
```
SELECT age_group,  
       SUM(purchase_amount) AS revenue  
FROM customer  
GROUP BY age_group  
ORDER BY revenue DESC;
```

age_group	revenue
young-adult	62,143
middle_aged	59,197
adult	55,978
senior	55,763

Young adults bring in the most revenue (27%), but the gap between groups is small – all ages matter.

5 Power BI Dashboard

An interactive dashboard was built in Power BI so managers can explore the data themselves. Filters at the top (**Category, Location, Age Group, Payment Method**) update every visual at once.



KPI cards (top row)

Card	Value
Avg Review Rating	3.75
Avg Previous Purchases	25.35
Avg Purchase Amount	\$59.76
Avg Purchase Days	89.13
Number of Customers	3.9K
Total Purchase Amount	\$233K

Visuals

- **Bar chart** – revenue by category
- **Column chart** – customers by age group
- **Donut chart** – customers by gender
- **Map** – customers by US state
- **Column chart** – revenue by season
- **Treemap** – number of purchases by category (with drill-down)
- **Decomposition tree** – total revenue broken down by category → age group → colour

Main DAX measures

```
Total Purchase Amount = SUM(customer[purchase_amount])
Avg Purchase Amount = AVERAGE(customer[purchase_amount])
Avg Review Rating = AVERAGE(customer[review_rating])
Avg Previous Purchases = AVERAGE(customer[previous_purchases])
Avg Purchase Days = AVERAGE(customer[purchase_frequency_days])
Number of Customers = DISTINCTCOUNT(customer[customer_id])
```

Pages: Customer Profile · Purchase Patterns · Product Preferences. The decomposition tree lets users click step by step to see which category, age group and colour drive revenue (e.g. Clothing → young-adult → Maroon).

6 Business Recommendations

1. Open the subscription to women - and make it worth joining

All 1,053 subscribers are men; 0 of 1,248 women subscribe. Subscribers also spend the same per order as everyone else.

- Run a subscription campaign aimed at female customers (Clothing and Accessories offers).
- Add real member benefits: free Express shipping, early access to new items, member-only prices.

Why: if just 20% of women subscribe, that is ~250 new subscribers.

2. Convert loyal non-subscribers

2,583 customers with 5+ previous purchases are not subscribed.

- Send them a personal “you already shop with us often – join and save” offer.
- Use points / rewards for every purchase so loyalty is visible.

Why: these customers already trust the brand; they are the cheapest to convert.

3. Use discounts smarter

43% of orders use a discount, but discounts do not increase order size (\$59.28 vs \$60.13). Hats, Coats and Sneakers are sold with a discount about half the time.

- Replace blanket discounts with “spend \$80, get 10% off” style offers that grow the basket.
- Reduce discounting on Hats, Coats and Sneakers and test the effect on sales.

Why: protects profit margin without losing customers.

4. Focus stock and marketing on hero products

- Keep Jewelry, Blouse, Pants, Shirt, Sandals and Jacket always in stock and well promoted.
- Bundle best-sellers with high-rated items (Gloves, Sandals, Boots) – e.g. “Jacket + Gloves”.
- Grow Outerwear (only 8% of revenue) with Fall/Winter campaigns.

5. Promote faster shipping and fix the Summer dip

- Express / 2-Day customers spend ~\$2 more – offer free Express above a minimum order.
- Summer is the weakest season (–7% vs Fall): plan a summer sale on Sandals, Hats and Accessories.

6. Attract new customers and improve ratings

- Only 4% of customers are new – invest in first-purchase offers and referral rewards (“bring a friend”).
- Average rating is only 3.75/5: collect feedback on low-rated items and fix quality or sizing issues.

Conclusion: the business has a strong, loyal customer base and steady sales across seasons, states and payment methods. The biggest opportunities are not in finding new products, but in **making the subscription attractive (especially to women), using discounts to grow basket size**, and **bringing in new customers**.